

AIT503 Introduction to Optimization

PROPOSAL PROJECT

Dataset: IPL ball-by-ball dataset

By - Sriram Srikanth (IMT2023115)

ABOUT THE DATASET

Sources:

<https://www.kaggle.com/datasets/patrickb1912/ipl-complete-dataset-20082020>

<https://github.com/ritesh-ojha/IPL-DATASET>

Number of data points: **278206**

Number of input features: **14**

Number of unconditional output labels: **3**

Number of conditional output labels: **2**



INPUT FEATURES

Feature Name	Data Type	What it means ?
id	int	Uniquely identifies a match in IPL
Innings	int	The current innings number (1 or 2, sometimes 3/4 in special cases)
Overs	int	The over number within the innings (0–20 in T20)
BallNumber	int	Ball number within the over (1–6, extra balls possible)

INPUT FEATURES

Feature Name	Data Type	What it means ?
Batter	string	Name of the batter on strike
Bowler	string	Name of the bowler delivering the ball
NonStriker	string	Name of the batter at the non-striker's end
FieldersInvolved	string	Name(s) of fielder(s) involved for the ball (separated by ,)
BattingTeam	string	Team name of the batting side
BowlingTeam	string	Team name of the bowling side

INPUT FEATURES

Feature Name	Data Type	What it means ?
city	string	City where the match is played
venue	string	Stadium/venue name
umpire1	string	Name of the first on-field umpire
umpire2	string	Name of the second on-field umpire

OUTPUT LABELS

Unconditional Labels: Their need to be predicted does not depend on other output labels

Label Name	Data Type	What it means ?
BatsmanRun	int	Runs scored by the batter off the ball (0–6, includes boundaries and sixes)
ExtrasRun	int	Runs given as extras (wides, no-balls, leg-byes, byes, penalty runs)
PlayerOut	boolean (0/1)	Binary indicator: 1 if the ball resulted in a player getting out, 0 otherwise

OUTPUT LABELS

Conditional Labels: Their need to be predicted does not depend on other output labels. They provide more information on the variable they are dependent on.

Label Name	Data Type	What it means ?	Dependent On
ExtraType	string	Type of extra if any (e.g., Wide, No-ball, Leg-bye, Bye, Penalty); NULL/None if no extra	ExtraRuns
OutKind	string	Means by which batsman got out if any(e.g., Bowled, Caught, LBW, Run Out, Stumped, Hit Wicket, Retired Hurt, Obstructing Field); NULL/None if not out	PlayerOut

THE PROBLEM

To predict the ball-by-ball outcomes in an IPL match using historical data.

The aim is to predict:

- Runs scored by the batter.
- Extras runs recognized and their type.
- Whether the batsman gets out or not, and if so, the way in which the batsman got out.

WHY OPTIMISATION ?

Optimization of strategy, performance, and resources in high-pressure T20 matches

- **Complex Decision-Making:** To create an accurate predictive model, we need to apply optimization techniques where we systematically tune the model to minimise the error.
- **Limited Resources and Constraints:** captains have only 20 overs, 10 wickets, and limited bowler quotas how to maximize winning chances under constraints.
- **Analytics-driven decision-making:** reduces guesswork, provides data-backed and an algorithmic strategy

IMPACTS OF SOLUTION

Optimization of strategy, performance, and resources in high-pressure T20 matches

In cricket (specifically IPL), every ball can change the match outcome — runs scored, extras conceded, or wickets taken. For teams, coaches, and analysts, predicting ball-by-ball outcomes helps in:

- **Strategy optimization** (bowling changes, field placements, batting order decisions).
- **Performance analysis** (which bowler contains which batter best, high-risk deliveries, etc.).
- **Simulation & forecasting** (projecting total runs, win probabilities, over-by-over planning).

THANK YOU